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Adaptive Anisotropic Gaussian Splatting for Multi-contrast MRI Arbitrary-Scale Super-Resolution with Anatomy Guidance

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Abstract

Implicit neural representation (INR) based methods learn a continuous mapping from a low-resolution (LR) target magnetic resonance (MR) image and a high-resolution (HR) reference image to achieve arbitrary-scale super-resolution (SR). However, their inherent spectral bias favors learning low-frequency (LF) components, often failing to capture the sharp transitions at anatomical boundaries and resulting in the loss of high-frequency (HF) details. Inspired by 3D Gaussian splatting, we propose GaussM^{2}ASR (Gaussian Multi-contrast MRI Arbitrary-scale Super-Resolution), which converts the challenging task of HF anatomical reconstruction into a smoother parameter optimization problem by learning the parameters of anisotropic 2D Gaussian kernels. To handle inter-contrast discrepancies, we introduce an anatomy-guided pipeline comprising three core modules: a Structure Prior Modulation Fusion (SPMF) module for feature enhancement; an Anatomy-Guided Dual-Domain Cross Attention (AG-DDCA) module for joint spatial-frequency modeling; and an Anatomy-Guided Gaussian Parametrizer (AGGP) that leverages gradient-based sparse attention to concentrate Gaussian centers on critical anatomical structures. Extensive experiments on multiple datasets demonstrate that GaussM^{2}ASR surpasses state-of-the-art methods in recovering fine anatomical details. Our source codes have been released at https://github.com/Qiuhai-CV/GaussM2ASR.

1. Introduction

Magnetic resonance imaging (MRI) is widely used in clinical practice because it is noninvasive, free of ionizing radiation and provides high-resolution (HR) images. However, hardware constraints and long acquisition times make it difficult to directly obtain HR scans for all contrasts [22, 30, 34]. It is therefore attractive to use routinely acquired fast HR contrasts as guidance to perform super-resolution (SR) on slower contrasts. Although several deep learning-based multi-contrast MRI SR methods have been proposed [9, 18, 19, 21, 29, 38], these methods typically operate at a fixed scaling factor, limiting their applicability in clinical workflows that require arbitrary scaling.

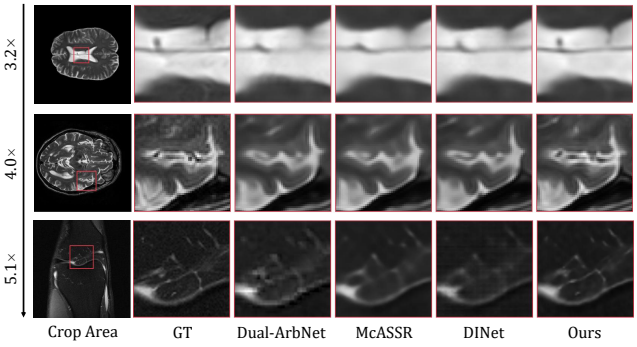


Figure 1. The proposed GaussM^{2}ASR recovers richer anatomical boundaries than INR-based models, e.g., Dual-ArbNet [41], McASSR [16], and DINet [36].

Recently, Li et al. [16] proposed an implicit neural representations (INR)-based arbitrary-scale SR method for multi-contrast MRI by learning a continuous mapping from a pair of low-resolution (LR) target and HR reference images to HR target images at arbitrary scales. Subsequent works further advanced this approach with scale-adaptive dynamic convolution [36] and scale-aware attention [43] to enhance model capacity. However, as illustrated in Fig. 1, these INR-based models often struggle to recover high-frequency (HF) details in HR target images, particularly along anatomical boundaries, thus limiting their ability to meet clinical demands for anatomical fidelity. As illustrated in Fig. 2(a), this limitation can be attributed to a fundamental **spectral bias**. This bias originates from the tendency of over-parameterized networks, under gradient-based optimization, to first converge towards low-frequency (LF) solution spaces, which correspond to functions of lower com-

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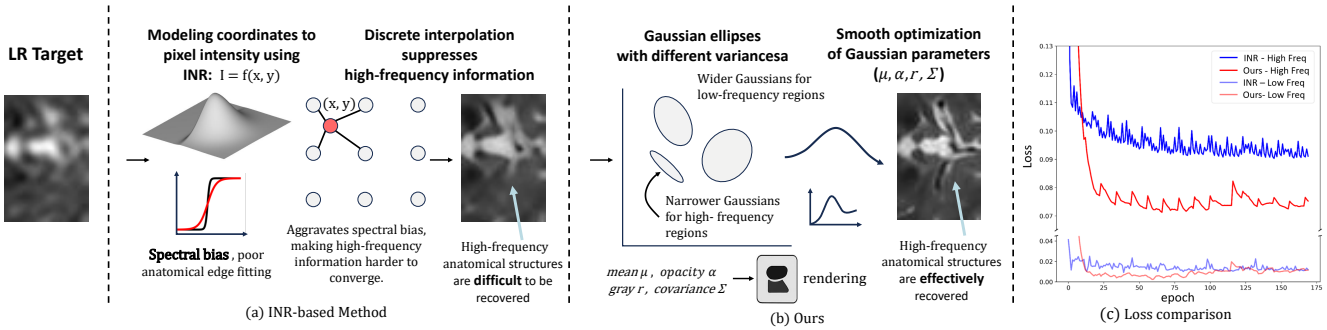


Figure 2. Compared with INR models that directly regress pixel intensities, our GaussM^{2}ASR learns a set of Gaussian parameters and reconstructs the image as a composition of adaptive basis functions, enabling more effective modeling of high-frequency anatomical structures.

plexity. Consequently, these models struggle to form the sharp spatial transitions required to represent HF anatomical structures, such as tissue boundaries. Furthermore, the continuous representation inherent to INR, $f : \mathbb{R}^2 \rightarrow \mathbb{R}$, often relies on an interpolation of a discrete latent feature grid. This interpolation acts as a low-pass filter, inherently attenuating HF components and thereby exacerbating the spectral bias.

In the domain of 3D scene reconstruction, 3D Gaussian splatting (3DGS) [7, 14, 26, 27, 32, 33, 40] has demonstrated superior efficiency and photorealistic rendering quality compared to INR approaches like NeRF. Inspired by this breakthrough, we introduce GaussM^{2}ASR (Gaussian Multi-contrast MRI Arbitrary-scale Super-Resolution), an arbitrary-scale SR framework for multi-contrast MRI that adapts the GS paradigm to a 2D imaging context. As illustrated in Fig. 2(b), our model leverages the inherent adaptability of anisotropic Gaussian kernels. These kernels dynamically adjust their variance: employing narrower kernels to capture HF details at anatomical boundaries, while utilizing wider kernels to represent LF homogeneous regions. Rather than directly regressing pixel intensities, our method learns a set of Gaussian parameters (mean, covariance, and opacity), effectively constructing the image as a composite of adaptable basis functions. This representation is inherently more suited to modeling HF content, thereby transforming the challenging task of HF recovery into a more tractable optimization problem over a smoother parameter space. A comparative analysis of the convergence behavior of an INR-based method (DINet [36]) and GaussM^{2}ASR is provided in Fig. 2(c). While both methods achieve comparable performance in fitting LF components, our 2DGS model demonstrates significantly accelerated and superior convergence in recovering HF structures, underscoring its efficacy in preserving critical anatomical details.

In the design of GaussM^{2}ASR for multi-contrast MRI SR, two key challenges arise. First, MR images often feature concentrated anatomical structures surrounded by large background areas with minimal texture, which can interfere with the SR process if not properly addressed. Second, it is crucial to effectively leverage the information from the HR reference image with varying contrast. To address these challenges, we propose an anatomy prior-driven [5] Gaussian modeling framework. We begin by using the Structure Prior Modulation Fusion (SPMF) module to enhance LR target features with statistical anatomical priors from the reference image, suppressing background interference. Next, the Anatomy Guided Dual-Domain Cross Attention (AG-DDCA) module enhances HF information in the frequency branch and aligns anatomical structures in the spatial branch, enabling the network to adaptively learn narrow kernels for HF and wide kernels for LF regions. Finally, the Anatomy Guided Gaussian Parametrizer (AGGP) module uses the reference image's anatomical gradient and a top-T sparse mask to guide Gaussian centers to focus on anatomical structures, ensuring sharper edges and more coherent fine structures.

Overall, our contributions are summarized as follows:

- Based on the analysis of the representation capabilities for HF detail information between INR and 2D Gaussian, we propose GaussM^{2}ASR, a 2DGS framework for multi-contrast MRI arbitrary-scale SR.
- We design three anatomy prior-driven modules—SPMF, AG-DDCA, and AGGP—that leverage statistical anatomical priors of the reference image to guide anisotropic Gaussian kernels in adapting to anatomical structures.
- Experiments on multiple datasets demonstrate that GaussM^{2}ASR outperforms state-of-the-art (SOTA) INR-based models in recovering fine anatomical details.

2. Related Work

2.1. Multi-Contrast MRI SR

MRI forms images by exciting water molecules in a magnetic field and capturing their relaxation signals. Its sensitivity to tissue properties and acquisition parameters allows

multiple contrasts—such as T1, T2, and PD-weighted images—each providing complementary diagnostic information. Therefore, using a fast HR reference image (e.g., T1) to enhance a slow LR target (e.g., T2) has become an important strategy to improve efficiency and harmonize data across different acquisitions.

Fixed-scale MRI SR. Early MRI SR studies focused on fixed integer scaling factor. MINet [8] proposes a multi-stage network that leverages the explicit dependencies between different contrasts to guide the SR process. The deep unfolding network MarDUN [35] was proposed by incorporating custom-designed gradient descent and proximal mapping modules. With the advent of Transformers [28], approaches like MTrans [10] further exploited cross-attention for reference–target feature fusion. Despite their success at specific scale factors (e.g., 2×, 4×), these methods lack flexibility, as each scale requires separate training and storage, limiting their practicality for arbitrary or non-integer scaling factors.

Arbitrary-scale MRI SR. To overcome the limitations of fixed-scale SR approaches, researchers have explored arbitrary-scale SR techniques, with INR emerging as a key solution. INR models an image as a continuous function mapping spatial coordinates to pixel intensities, typically realized through MLP layers, thus enabling reconstruction at arbitrary resolutions. In natural images, LIIF [6] pioneered this concept by learning a local implicit function from an LR feature grid, inspiring medical adaptations such as ArSSR [37] for 3D MRI and McASSR [16], DualArbNet [41] for multi-contrast MRI SR. However, INR-based methods still face challenges: their MLP-based per-pixel interpolation is computationally demanding and suffers from spectral bias, resulting in over-smooth reconstructions and the loss of HF structural details critical for clinical use.

2.2. Gaussian Splatting

GS has recently emerged as a novel scene representation technique that addresses the limitations of INR-based models in 3D modeling community. Originally introduced by Kerbl et al. [13] for real-time radiance field rendering, 3DGS represents a scene with numerous explicitly optimizable 3D Gaussians, each defined by position, covariance, color, and opacity. Unlike NeRF [23], which relies on implicit MLP-based coordinate mapping, this explicit representation enables real-time rendering, faster convergence, and superior preservation of fine geometry and HF details. Building on its success in 3D reconstruction, researchers have extended the concept to 2D tasks. For example, GaussianImage [42] applies 2D Gaussians for image compression, while 2DGS-inpaint [17] exploits their spatial continuity for image inpainting. More recently, methods such as ContinuousSR [25] and GSASR [4] have demonstrated that modeling pixels as continuous 2D Gaussian fields can effectively alleviate the over-smoothing problem inherent in INR-based methods by employing 2DGS for 2D modeling in the low-level domain.

3. Method

3.1. 2D Gaussian Splatting

To tackle the challenge of reconstructing HF details in MRI SR, the characteristics of 2DGS provide a potent solution. By capitalizing on the differentiability and composability of continuous Gaussian functions, this approach demonstrates a strong capacity for modeling fine-grained anatomical structures. The 2D Gaussian field at a spatial location $\mathbf{x}$ is defined by a probability density function parameterized by its mean $\boldsymbol{\mu}$ (position) and covariance matrix $\boldsymbol{\Sigma}$:

$$G(\mathbf{x}) = \frac{1}{2\pi|\boldsymbol{\Sigma}|^{1/2}} \exp\left(-\frac{1}{2}(\mathbf{x}-\boldsymbol{\mu})^\top \boldsymbol{\Sigma}^{-1}(\mathbf{x}-\boldsymbol{\mu})\right), \quad (1)$$

where $\boldsymbol{\mu} \in \mathbb{R}^2$ denotes the Gaussian mean, and $\boldsymbol{\Sigma} \in \mathbb{R}^{2\times 2}$ is a positive definite covariance matrix that determines the shape and orientation of the Gaussian. To characterize the anisotropic shape of each Gaussian, $\boldsymbol{\Sigma}$ is parameterized by two standard deviations, σ_x and σ_y, and a correlation coefficient ρ:

$$\Sigma = \begin{bmatrix} \sigma_x^2 & \rho\sigma_x\sigma_y \\ \rho\sigma_x\sigma_y & \sigma_y^2 \end{bmatrix}, \quad \sigma_x > 0,\ \sigma_y > 0,\ |\rho| < 1, \quad (2)$$

where σ_x and σ_y control the spatial spread along the two axes, and ρ captures their correlation.

During rasterization, the intensity $f(\mathbf{x})$ at a pixel is an alpha-blended composite of N overlapping Gaussians, each contributing a gray level r_i weighted by its opacity α_i and Gaussian evaluation $G_i(\mathbf{x})$:

$$f(\mathbf{x}) = \sum_{i=1}^{N} \alpha_i r_i G_i(\mathbf{x}). \quad (3)$$

3.2. GaussM^{2}ASR Framework

The overall pipeline of the proposed GaussM^{2}ASR framework is depicted in Fig. 3. The model takes as inputs an LR target image $I_{\text{tar}} \in \mathbb{R}^{h\times w\times 1}$ and an HR reference image $I_{\text{ref}} \in \mathbb{R}^{H\times W\times 1}$. It begins with a Cross-Scale Multi-contrast Fusion (CSMF) encoder to extract multi-scale deep features from both images. To effectively incorporate anatomical priors, these features are then restructured and enhanced by the **Structure Prior Modulation Fusion (SPMF)** module, which leverages the statistical anatomical information from I_{ref}. Thereafter, the **Anatomy Guided Dual-Domain Cross Attention (AG-DDCA)** module integrates scale embeddings with a learnable Gaussian prompt $P \in \mathbb{R}^{H\times W\times C}$ to perform complementary feature refinement in both spatial and frequency domains. A Gaussian

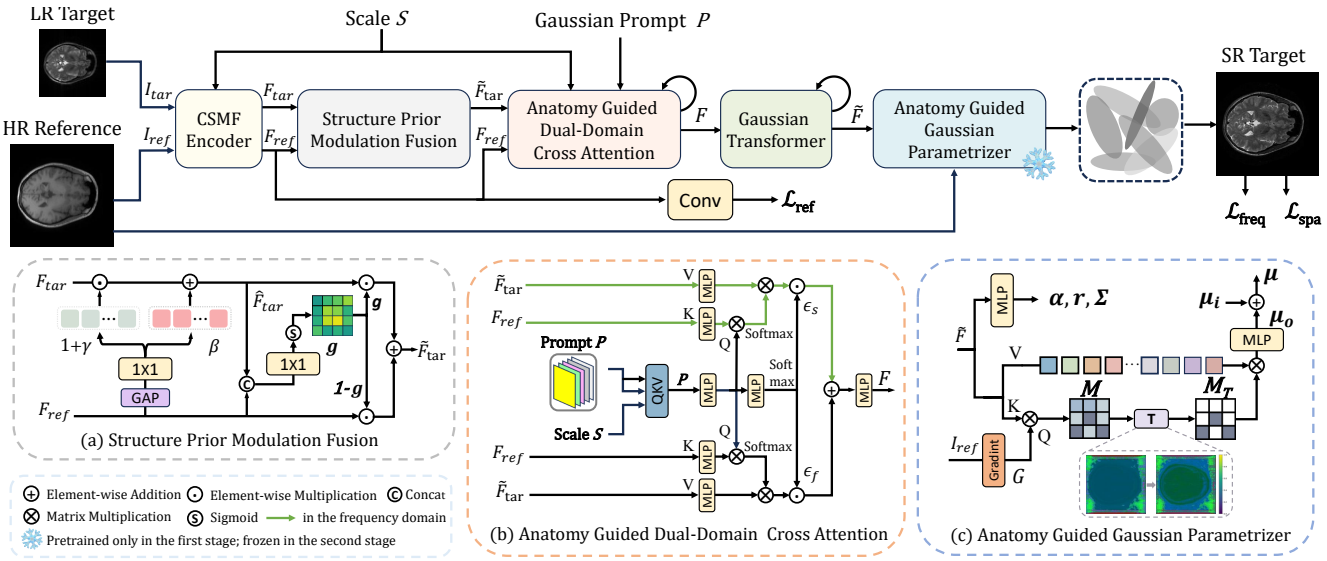


Figure 3. Overview of the proposed GaussM^{2}ASR, which adopts a two-stage training strategy: pre-training with HR target images, followed by fine-tuning with LR inputs while freezing the Anatomy Guided Gaussian Parametrizer (AGGP) module.

Transformer block is further employed to reinforce global coherence among the representations. In the final stage, the **Anatomy Guided Gaussian Parametrizer (AGGP)** estimates the Gaussian centers $\boldsymbol{\mu}$, while the other parameters are regressed via MLP heads. The resulting Gaussians are rendered in a continuous space, enabling the synthesis of high-fidelity SR images at arbitrary scaling factors. We will elaborate on the three most critical modules in detail. The specific architectures of CSMF encoder and Gaussian Transformer are described in the Supplementary Material.

3.2.1. Structure Prior Modulation Fusion

Direct fusion of target and reference features in multi-contrast MRI is challenged by two key issues: the prevalence of uninformative background regions and significant distribution shifts across contrasts. First, background areas lacking anatomical content can interfere with fusion. Second, inter-contrast statistical discrepancies may cause improper weighting of HF components, potentially obscuring structural details. To address these limitations, we propose a Structure Prior Modulation Fusion (SPMF) module after the CSMF encoder. Let F_{tar} and F_{ref} represent the features of the LR target and HR reference images, respectively. The module first computes channel-wise scaling and bias parameters (γ, β) from the global statistics of the reference features

$$(\gamma, \beta) = \Phi(\text{GAP}(F_{\text{ref}})), \tag{4}$$

$$\hat{F}_{\text{tar}} = (1 + \gamma) \odot F_{\text{tar}} + \beta, \tag{5}$$

where $\Phi(\cdot)$ is a 1×1 convolution followed by ReLU, $\odot$ is element-wise multiplication, and $\text{GAP}(\cdot)$ denotes global average pooling. This operation enhances channels encoding HF structural information through feature-wise affine transformation. A pixel-wise gating mechanism then performs spatial refinement:

$$g = \text{Sigmoid}(\Phi(\text{Concat}[\hat{F}_{\text{tar}}, F_{\text{ref}}])), \tag{6}$$

$$\tilde{F}_{\text{tar}} = g \odot \hat{F}_{\text{tar}} + (1 - g) \odot F_{\text{ref}}, \tag{7}$$

where $\text{Cat}[\cdot, \cdot]$ is channel-wise concatenation. The gate g preserves $\hat{F}_{\text{tar}}$ in structurally salient regions (e.g., edges and textures), while relying on F_{ref} in less informative areas, thereby suppressing background interference.

3.2.2. Anatomy Guided Dual-Domain Cross Attention

When fusing target and reference features solely in the spatial domain, it is often challenging to effectively capture HF components that are critical for clinical diagnosis in MRI, such as tissue boundaries and fine anatomical details. This limitation not only restricts the effective utilization of reference priors but also tends to result in over-smoothed reconstructions.

To overcome this challenge, we propose an Anatomy Guided Dual-Domain Cross Attention (AG-DDCA) module, as illustrated in Fig. 3(b). The module introduces a learnable Gaussian prompt P and incorporates scale-aware cross-attention through a scale embedding S, enabling the Gaussian prompt to adapt to different scaling factors. The refined Gaussian prompt serves as the query Q for dual-domain cross-attention: the spatial branch captures global context from both target and reference features, while the frequency branch, operating on Fourier magnitude representations, emphasizes HF components to enhance edges and fine details. The attention mechanisms in both domains

are formulated as:

$$\text{Att}_{\text{spat}} = \text{Softmax}(QK_s^\top + B)V_s, \tag{8}$$

$$\text{Att}_{\text{freq}} = \text{Softmax}(QK_f^\top + B)V_f, \tag{9}$$

where K_s and V_s denote the keys and values from spatial-domain features of the reference and target images, while K_f and V_f represent their frequency-domain counterparts. The term B denotes the relative position bias.

The outputs from both domains are adaptively integrated through a dynamic gating mechanism conditioned on the Gaussian prompt:

$$F = \epsilon_s \odot \text{Att}_{\text{spat}} + \epsilon_f \odot \text{Att}_{\text{freq}}, \tag{10}$$

where the gating weights ϵ_s and ϵ_f are obtained by processing the Gaussian prompt P through a two-layer MLP followed by a Softmax activation.

3.2.3. Anatomy Guided Gaussian Parametrizer

In continuous Gaussian representations, the spatial support and influence of each Gaussian are defined by its center. Misalignment between these centers and true anatomical structures—particularly at tissue boundaries and fine details—often forces the model to compensate by introducing wider Gaussians. Such adjustments, however, tend to oversmooth critical edges and diminish HF information.

To mitigate this, we propose an Anatomy Guided Gaussian Parametrizer (AGGP). It first computes a gradient map $G = \nabla I_{\text{ref}}$ from the reference image via the Sobel operator. This map is fused with the feature representation $\tilde{F}$ from the Gaussian Transformer using a sparse top-T cross-attention mechanism, which suppresses responses from homogeneous regions while preserving channels correlated with anatomical edges. The mechanism is formulated as:

$$W = \frac{QK^\top}{\tau}, \tag{11}$$

$$M_{ij} = \begin{cases} 1, & \text{if } j \in \text{Top}_T(W_{i,:}), \\ 0, & \text{otherwise}, \end{cases} \tag{12}$$

$$\text{Att}_{\text{top-T}}(Q, K, V) = \text{Softmax}\left(M \odot W\right) V, \tag{13}$$

where $Q = G$, and K, V are projections of $\tilde{F}$. Here, $W \in \mathbb{R}^{C\times C}$ is the attention score matrix, and $\text{Top}_T(\cdot)$ selects the top-T values per row to construct the binary mask M.

A three-layer MLP then estimates a center offset $\boldsymbol{\mu}_o$ from the attended features. The final Gaussian center $\boldsymbol{\mu} = \boldsymbol{\mu}_i + \boldsymbol{\mu}_o$ is obtained by adding $\boldsymbol{\mu}_o$ to a uniformly initialized position $\boldsymbol{\mu}_i$, ensuring spatial coverage while adapting to significant regions. The remaining parameters (covariance $\boldsymbol{\Sigma}$, opacity $\boldsymbol{\alpha}$, intensity $\boldsymbol{r}$) are regressed from $\tilde{F}$ via lightweight MLP heads. All parameters are used for 2DGS (Sec. 3.1) to generate a high-fidelity HR image with well-preserved structural details.

3.3. Training and Optimization

The training objective combines multiple loss terms to regularize the optimization process:

$$\mathcal{L}_{\text{total}} = \mathcal{L}_{\text{spa}} + \lambda_{\text{freq}}\mathcal{L}_{\text{freq}} + \lambda_{\text{ref}}\mathcal{L}_{\text{ref}}, \tag{14}$$

where $\lambda_{\text{freq}} = 0.01$ and $\lambda_{\text{ref}} = 0.3$ are empirically determined weighting coefficients that balance the contribution of each loss component. The spatial loss $\mathcal{L}_{\text{spa}}$ minimizes the mean absolute error (MAE) between the GT and SR target images in the spatial domain, ensuring pixel-level fidelity. The frequency loss $\mathcal{L}_{\text{freq}}$ operates in the Fourier domain to preserve HF components by reducing the MAE between the magnitude spectra of ground-truth (GT) and SR images. The reference loss $\mathcal{L}_{\text{ref}}$ maintains consistency between the input HR reference image and its reconstruction from the reference feature embedding, thereby stabilizing the cross-contrast prior transfer process.

4. Experiment and Analysis

4.1. Experiment Setting

Datasets. We conduct experiments on three public multi-contrast MRI datasets, including IXI, BraTS [3], and fastMRI [39]. Details of the datasets are summarized in Table 1. For each volumetric data, we used a center-cropping method to extract the middle 15% slices as training data. During both training and testing, GT images (256×256) are downsampled at a scale factor $S \in (1, 4]$ via k-space cropping [15] to generate LR target images. The multi-contrast MRI scans from fastMRI and BraTS are pre-aligned to a common anatomical template. For the IXI dataset, we perform pre-registration of the multi-contrast scans using FSL-FLIRT, in accordance with previous works [12].

Table 1. Details of three datasets used in experiments.

Datasets	IXI	BraTS	fastMRI
Reference	T1	T1	FD
Target	T2	T2	FSPD
Train / Test	500 / 77	600 / 200	500 / 53

Implementation details. Our GaussM^{2}ASR is implemented in `PyTorch` [24] and trained on a single NVIDIA RTX A100 GPU (40 GB). We use the Adam [1] optimizer (β_1=0.9, β_2=0.99) with an initial learning rate of 2×10^{-4}, which is decayed by a MultiStepLR scheduler at 250k, 400k, 450k, and 475k iterations. Training runs for 500k iterations with a batch size of 1, stabilized by an Exponential Moving Average (EMA) decay rate of 0.999. We adopt a two-stage strategy: pre-training with HR target images, followed by fine-tuning with LR inputs while freezing the AGGP module.

The CSMF encoder comprises 16 residual blocks (64 channels). Subsequent modules (from SPMF onward) operate in a 180-dimensional embedding space with each attention layer utilizing 6 attention heads. The AG-DDCA uses a single cross-attention layer, while the Gaussian Transformer employs 5 self-attention layers for feature refinement. The AGGP uses four sparse attention branches with sparsity ratios of $\{1/2, 2/3, 3/4, 4/5\}$. For the covariance parameterization, σ_x and σ_y use scaled sigmoid constraints, while ρ uses tanh; α is constrained by a scaled sigmoid function. The model allocates one Gaussian per reference image pixel, with density modulated by the reference resolution to support arbitrary scaling factors.

4.2. Comparative Methods and Evaluation Metrics

To comprehensively evaluate GaussM^{2}ASR, we compare it against a range of SOTA methods, which are categorized into two groups: 1) Models originally designed for natural image SR, including LIIF [6], GaussianSR [11], and GSASR [4]. For a fair comparison in the multi-contrast MRI, we adapt these methods by using EDSR [20] as the backbone and concatenating the reference and target images as input for arbitrary-scale SR. 2) Methods specifically developed for multi-contrast MRI SR, consisting of Dual-ArbNet [41], McASSR [16], and DINet [36]. All competing methods were retrained on our experimental datasets using their official implementations to ensure a fair comparison.

We report performance on in-scale factors within the training distribution (1.5×, 2×, 3×, and 4×). To further test the generalization ability of the models, we also evaluate on out-of-scale factors (5× and 6×) that were not seen during training. Quantitative assessment is performed using two standard image quality metrics: the Peak Signal-to-Noise Ratio (PSNR) [2] and the Structural Similarity Index Measure (SSIM) [31].

4.3. Quantitative Results

The quantitative results of GaussM^{2}ASR and baseline methods are presented in Table 2. It reveals that our GaussM^{2}ASR achieves dominant performance, which consistently ranks at the top across both in-scale and out-of-scale SR tasks, demonstrating remarkable robustness. Notably, our model attains a significantly higher SSIM than all competing approaches. This advantage indicates that GaussM^{2}ASR excels at faithfully reconstructing anatomical structures and preserving HF information across arbitrary scaling factors, ensuring high-fidelity representations.

4.4. Qualitative results

Fig. 4 presents qualitative comparisons of the proposed GaussM^{2}ASR with baseline methods on three datasets at a scaling factor of 4. INR-based models, such as LIIF [6] and Dual-ArbNet [41], tend to produce over-smoothed reconstructions, a phenomenon stemming from spectral bias. In contrast, GaussM^{2}ASR reconstructs finer anatomical details and sharper tissue boundaries, owing to its use of anisotropic Gaussian kernels that effectively enhance HF regions and maintain structural continuity. Furthermore, when compared to GS-based methods like GaussianSR [11] and GSASR [4], GaussM^{2}ASR preserves richer HF textures and exhibits sharper tissue transitions. This is primarily because our model effectively leverages anatomical priors from the reference images.

We further provide error maps between the SR results and the GT image for each method. These maps demonstrate that GaussM^{2}ASR achieves lower reconstruction errors, particularly in regions with fine anatomical details. The visual evidence confirms that our method not only achieves superior perceptual fidelity but also more accurately preserves diagnostically critical structures.

To elucidate the high-fidelity modeling mechanism of GaussM^{2}ASR, Fig. 5 visualizes the spatial distribution and shapes of the learned Gaussian kernels. The results show that our method dynamically allocates narrower kernels to HF regions, while employing wider kernels in smoother, LF areas. Furthermore, the Gaussian centers are densely concentrated along anatomical structures and sparse in background regions. This visualization quantitatively validates that our approach adaptively configures Gaussians to prioritize anatomically significant areas, which is the foundation of its superior reconstruction performance.

4.5. Ablation Study

In this section, we present ablation comparisons on the model architecture and training strategies, with the metrics of 4× SR task on the IXI dataset summarized in Fig. 7. Additional analysis on model architecture and runtime efficiency, and further visual results can be found in the supplementary material.

Analysis on the SPMF module. Removing the SPMF module (w/o SPMF) leads to a notable performance decline. This is primarily because background regions in MR images often introduce distracting activations that interfere with feature fusion. As illustrated in Fig. 6, integrating structural priors from the reference image allows the model to reweight the target feature channels and spatially modulate the LR input. This process effectively attenuates irrelevant background responses while accentuating anatomy-related.

Analysis on the AG-DDCA module. We investigate the importance of the frequency branch in AG-DDCA by removing it and performing cross-attention only in the spatial domain (w/o freq). The resulting drop in both PSNR and SSIM underscores the effectiveness of the dual-domain operation. This verifies that the frequency branch provides complementary information to the spatial features, which is essential for the accurate recovery of image details.

Table 2. Quantitative comparison on IXI, BraTS, and fastMRI datasets under different scaling factors. Red indicates the best and blue the second best performance.

Dataset	Method	in-scale								out-of-scale			
		1.5×		2×		3×		4×		5×		6×	
		PSNR	SSIM	PSNR	SSIM	PSNR	SSIM	PSNR	SSIM	PSNR	SSIM	PSNR	SSIM
IXI	GaussianSR [11]	33.47	0.9419	33.45	0.9429	29.80	0.8886	27.75	0.8479	26.04	0.8041	24.79	0.7637
	GSASR [4]	36.57	0.9487	34.28	0.9459	30.33	0.8950	28.38	0.8540	26.36	0.8144	25.08	0.7728
	LIIF [6]	37.73	0.9634	35.78	0.9538	31.26	0.9136	28.98	0.8768	27.01	0.8336	25.46	0.7955
	Dual-ArbNet [41]	41.54	0.9846	36.95	0.9619	32.50	0.9245	29.97	0.8903	27.75	0.8500	26.16	0.8124
	McASSR [16]	42.33	0.9802	38.32	0.9662	32.84	0.9298	30.21	0.8992	27.97	0.8527	26.37	0.8116
	DINet [36]	42.81	0.9864	38.43	0.9699	33.54	0.9367	30.98	0.9084	28.33	0.8684	26.58	0.8261
	Ours	46.62	0.9939	38.77	0.9798	34.25	0.9565	32.03	0.9350	28.81	0.8745	26.89	0.8295
BraTS	GaussianSR [11]	31.95	0.9864	34.33	0.9835	34.30	0.9687	32.45	0.9511	30.54	0.9301	29.08	0.9118
	GSASR [4]	38.04	0.9870	36.33	0.9841	34.11	0.9663	31.99	0.9474	30.04	0.9255	28.70	0.9072
	LIIF [6]	38.37	0.9882	37.83	0.9805	34.39	0.9591	31.94	0.9349	30.71	0.9279	29.35	0.9114
	Dual-ArbNet [41]	41.51	0.9939	39.22	0.9870	35.49	0.9698	32.97	0.9510	30.85	0.9300	29.27	0.9118
	McASSR [16]	38.98	0.9922	39.47	0.9816	35.12	0.9655	32.88	0.9481	30.68	0.9289	29.26	0.9129
	DINet [36]	39.29	0.9941	40.03	0.9878	35.86	0.9715	33.32	0.9543	31.18	0.9337	29.54	0.9143
	Ours	45.39	0.9959	41.22	0.9901	37.14	0.9769	34.65	0.9621	31.63	0.9444	29.85	0.9225
fastMRI	GaussianSR [11]	30.84	0.8487	30.13	0.8009	29.87	0.7377	27.66	0.7002	24.29	0.6683	23.98	0.6485
	GSASR [4]	32.14	0.8617	30.97	0.8054	30.02	0.7421	27.98	0.7084	25.47	0.6745	24.67	0.6546
	LIIF [6]	32.48	0.8654	31.28	0.8088	30.17	0.7453	28.24	0.7106	26.14	0.6787	25.32	0.6557
	Dual-ArbNet [41]	32.51	0.8688	31.65	0.8095	30.23	0.7453	28.59	0.7218	26.72	0.6912	26.06	0.6566
	McASSR [16]	32.64	0.8708	31.99	0.8116	30.49	0.7498	28.43	0.7143	26.55	0.6911	25.64	0.6617
	DINet [36]	32.73	0.8724	32.17	0.8124	30.74	0.7512	28.76	0.7164	26.68	0.6952	25.93	0.6672
	Ours	34.91	0.8832	33.18	0.8275	31.64	0.7733	30.53	0.7410	28.11	0.7008	26.69	0.6728

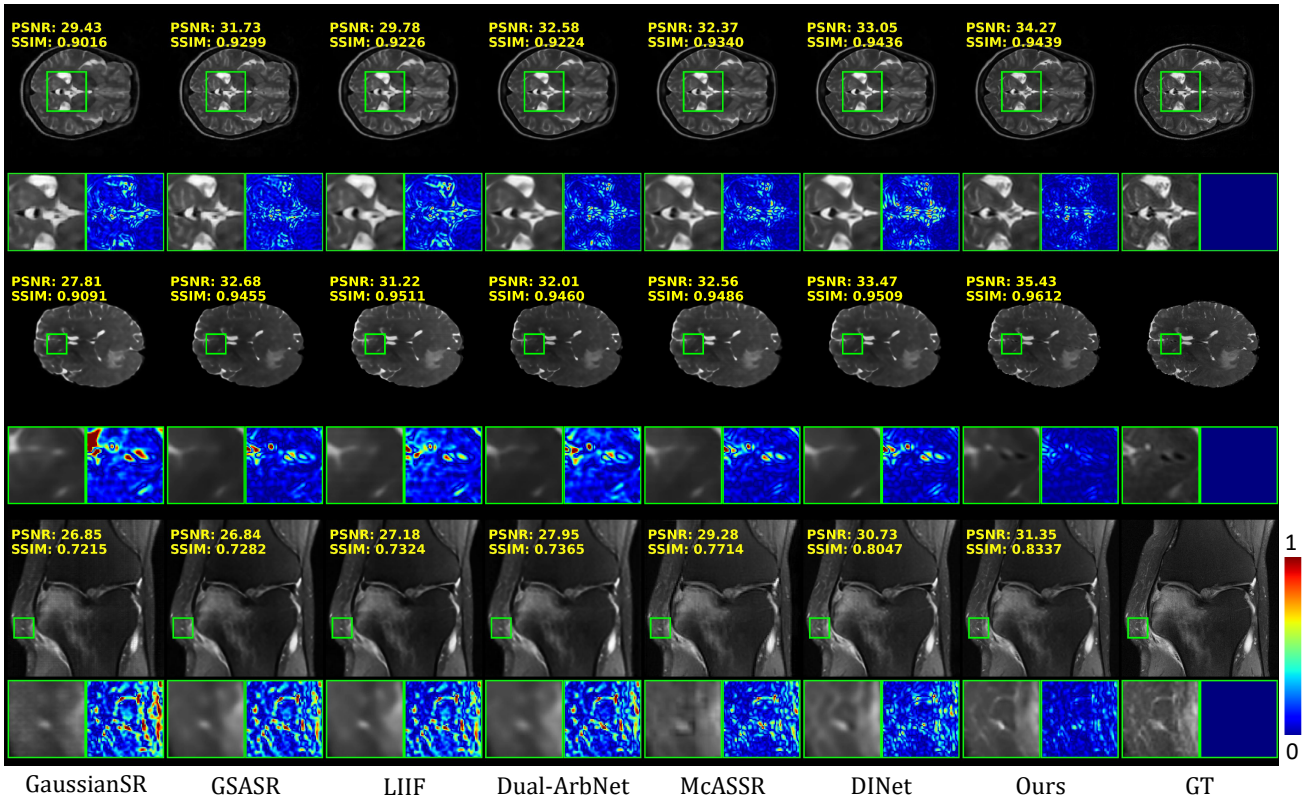
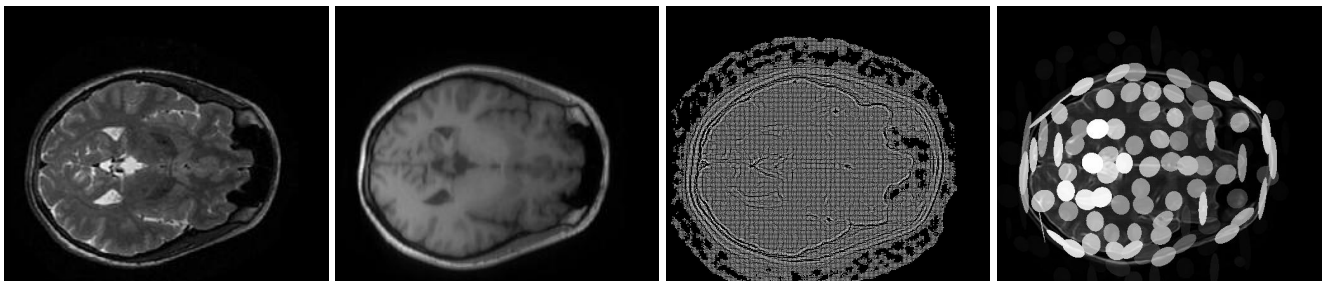


Figure 4. Qualitative comparison on IXI, BraTS, and fastMRI datasets at a 4× scaling factor. Enlarged views of regions within green boxes are provided alongside their error maps, which display the absolute reconstruction error (darker shades indicate smaller errors).

Analysis on the AGGP module. Ablating the Top-T sparse attention mechanism (w/o Top-T) reveals its critical role in guiding Gaussian centers toward anatomical boundaries, which is essential for producing sharper edges and more coherent fine structures; visual comparisons are provided in the supplementary material. Moreover, removing



(a) SR Target (b) HR Reference (c) Position Distribution (d) Visual Gaussians

Figure 5. Illustration of how the generated Gaussians render the MRI image, by showing their spatial distribution in (c) and a random subset in (d). The distribution in (c) concentrates on HF regions, particularly along anatomical boundaries, while the subset in (d) demonstrates the adaptive adjustment of their variances to local frequency characteristics.



(a) Before SPMF (b) After SPMF (c) Residual Map

Figure 6. Feature responses before and after the SPMF module, along with their residual map. The results demonstrate the module's efficacy in suppressing activations in background and other irrelevant regions.

the uniformly initialized positions (w/o $\boldsymbol{\mu}_i$) and directly predicting the Gaussian coordinates (i.e., setting $\boldsymbol{\mu} = \boldsymbol{\mu}_o$) enlarges the optimization space, leading to ineffective convergence of the Gaussian centers.

Analysis on the two-stage training strategy. The proposed two-stage training strategy involves pre-training the AGGP module on HR data to initialize anatomically-aware Gaussians, followed by freezing AGGP and fine-tuning the network on LR inputs. Conversely, a single-stage strategy directly trains the entire model on LR data. Our experiments reveal that the single-stage approach leads to a significant performance decline. The superior performance of the two-stage paradigm confirms its effectiveness in enhancing the model's ability to fit MR images, establishing it as a more advantageous training scheme for Gaussian-based representations.

5. Limitation and Future Work

It is worth noting that GaussM^{2}ASR requires the target and reference images to be spatially aligned during both training and inference. In clinical practice, however, patient movement may introduce misalignment and motion artifacts. Consequently, multi-contrast image registration is employed as a necessary pre-processing step. Additionally, the number of Gaussian kernels in our model is fixed and determined by the reference image resolution. For images

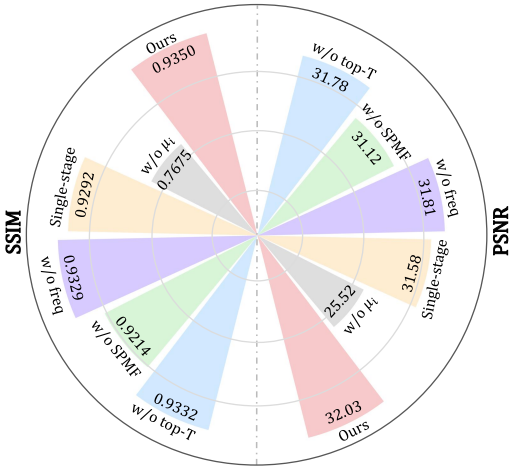


Figure 7. Ablation study of GaussM^{2}ASR, with all models trained on the IXI dataset and tested at a 4× upscaling factor.

with simpler textures, this can lead to redundant computations. To address these limitations, our future work will focus on integrating robust motion correction techniques and developing adaptive Gaussian allocation strategies to enhance the model's practicality and computational efficiency.

6. Conclusion

This paper presents GaussM^{2}ASR, a 2DGS–based model for arbitrary-scale multi-contrast MRI SR. It addresses the spectral bias of INR-based methods by adaptively modeling images with anisotropic Gaussian kernels—using narrow kernels for HF details and wide kernels for smooth areas. Through our carefully designed modules, the model effectively suppresses background interference, enhances anatomical features in dual domains, and precisely aligns Gaussian centers with critical structures. Extensive experiments confirm that GaussM^{2}ASR sets a new state-of-the-art, delivering superior SR fidelity with exceptionally well-preserved anatomical boundaries.

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